Norms and Metrics

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1 Introduction

A vector space gives us addition and scalar multiplication, but no notion of *length* or *distance*. The moment we want to talk about convergence of an iterative algorithm, continuity of a loss function, or generalisation bounds on a hypothesis class, we need to measure how close two vectors are. *Norms* and *metrics* provide exactly this quantitative geometry.

The two notions are tightly related: every norm on a vector space induces a metric in a canonical way. The converse fails, and the failure is instructive — it shows that metrics are strictly more general than norms, and that some legitimate measures of distance (e.g. the discrete metric, or the Kullback–Leibler divergence after suitable repair) cannot be derived from any notion of vector length.

2 Definitions

First, an example

Before stating axioms, let us measure one concrete vector three different ways. Take $v = (3, -4, 12) \in \mathbb{R}^3$. The ℓ^1 norm sums absolute entries: $\|v\|_1 = |3| + |-4| + |12| = 3 + 4 + 12 = 19$. The ℓ^2 (Euclidean) norm takes the square root of the sum of squares: $\|v\|_2 = \sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. The ℓ^∞ norm picks the largest absolute entry: $\|v\|_\infty = \max(3, 4, 12) = 12$. Now check the triangle inequality under ℓ^2 on the pair $x = (3, 0, 0)$ and $y = (0, 4, 0)$: we have $\|x\|_2 = 3$, $\|y\|_2 = 4$, and $x + y = (3, 4, 0)$ so $\|x + y\|_2 = \sqrt{9 + 16} = 5 \leq 3 + 4 = 7$. Good. The same vector therefore has length 19, 13, or 12 depending on which norm we use — a vivid reminder that "length" is a choice, not a given. The axioms below pin down exactly which choices are legitimate.

Throughout, let $\mathbb{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let V be a vector space over $\mathbb{F}$.

Definition 1 (Norm)**.** A function $\|\cdot\|\colon V \to \mathbb{R}$ is a *norm* if for every $x, y \in V$ and every scalar $\alpha \in \mathbb{F}$,

1. (Positive definiteness) $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$.
2. (Absolute homogeneity) $\|\alpha x\| = |\alpha| \, \|x\|$.
3. (Triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$.

The pair $(V, \|\cdot\|)$ is called a *normed vector space.*

Definition 2 (Metric)**.** Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is a *metric* (or *distance function*) if for every $x, y, z \in X$,

1. (Non-negativity) $d(x, y) \geq 0$.
2. (Identity of indiscernibles) $d(x, y) = 0 \iff x = y$.
3. (Symmetry) $d(x, y) = d(y, x)$.
4. (Triangle inequality) $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is called a *metric space.*

Definition 3 (Induced metric)**.** Let $(V, \|\cdot\|)$ be a normed vector space. The *metric induced by* $\|\cdot\|$ is

$$d(x, y) := \|x - y\|, \qquad x, y \in V.$$

Example 1 (ℓ^p norms on $\mathbb{R}^n$)**.** For $1 \leq p < \infty$ and $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, define

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}, \qquad \|x\|_\infty := \max_{1 \leq i \leq n} |x_i|.$$

Read aloud: the ℓ^p norm of x is defined as the p-th root of the sum from i equals one to n of the absolute value of x sub i raised to the p; and the ℓ-infinity norm of x is the maximum, over indices i from one to n, of the absolute value of x sub i. Each $\|\cdot\|_p$ is a norm. The triangle inequality for $p \geq 1$ is the *Minkowski inequality.*

3 Main theorem

Theorem 1 (Norms induce metrics)**.** *Let $(V, \|\cdot\|)$ be a normed vector space. Then the function*

$$d\colon V \times V \to \mathbb{R}, \qquad d(x, y) := \|x - y\|,$$

is a metric on V. Read aloud: *d is a function from the Cartesian product V cross V into the real numbers, and d of x comma y is defined as the norm of x minus y.*

Proof. We verify the four axioms of a metric.

(1) Non-negativity. By positive definiteness of the norm, $d(x, y) = \|x-y\| \geq 0$ for all $x, y \in V$.

(2) Identity of indiscernibles. We use positive definiteness again:

$$d(x, y) = 0 \iff \|x - y\| = 0 \iff x - y = 0 \iff x = y.$$

Read aloud: d of x comma y equals zero if and only if the norm of x minus y equals zero, if and only if x minus y equals zero, if and only if x equals y.

(3) Symmetry. By absolute homogeneity with $\alpha = -1$,

$$d(x, y) = \|x - y\| = \big\|(-1)\,(y - x)\big\| = |-1|\,\|y - x\| = \|y - x\| = d(y, x).$$

Read aloud: d of x comma y equals the norm of x minus y, which equals the norm of negative one times the quantity y minus x, which by absolute homogeneity equals the absolute value of negative one times the norm of y minus x, which is just the norm of y minus x, which is d of y comma x.

(4) Triangle inequality. For any $x, y, z \in V$, write $x - z = (x - y) + (y - z)$. The triangle inequality for the norm gives

$$d(x, z) = \|x-z\| = \big\|(x-y)+(y-z)\big\| \leq \|x-y\|+\|y-z\| = d(x, y)+d(y, z).$$

Read aloud: d of x comma z equals the norm of x minus z, which equals the norm of the sum x minus y plus y minus z, which by the triangle inequality is at most the norm of x minus y plus the norm of y minus z, equal to d of x comma y plus d of y comma z.

All four axioms hold, so d is a metric on V. □

4 A metric that is not norm-induced

Proposition 2 (Discrete metric)**.** *Let V be a nonzero vector space and define*

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

Read aloud: *d of x comma y equals zero whenever x equals y, and equals one whenever x is not equal to y. Then d is a metric on V, but there is no norm $\|\cdot\|$ on V such that $d(x,y) = \|x-y\|$ for all x, y.*

Proof. That d is a metric is a direct check of the four axioms.

Suppose for contradiction that $d(x,y) = \|x-y\|$ for some norm $\|\cdot\|$. Pick any $x \neq 0$. By homogeneity,

$$\|2x\| = 2\,\|x\|.$$

Read aloud: the norm of two x equals two times the norm of x. On the other hand, $2x \neq 0$ and $x \neq 0$, so

$$\|2x\| = d(2x, 0) = 1 \qquad \text{and} \qquad \|x\| = d(x,0) = 1.$$

Read aloud: the norm of two x equals d of two x comma zero, which equals one; and the norm of x equals d of x comma zero, which also equals one. Combining the displays gives $1 = 2$, a contradiction. Hence d does not come from any norm. □

5 Why this matters

Once a metric is fixed, the entire machinery of analysis is unlocked: open and closed sets, continuity, convergence, Cauchy sequences, completeness, compactness, Lipschitz constants. In machine learning, the choice of norm is rarely innocent: ℓ^1 regularisation produces sparse solutions, ℓ^2 regularisation shrinks coefficients isotropically, and the spectral (operator) norm controls how much a neural-network layer can amplify inputs. The Wasserstein distance is a genuine metric on probability distributions and underlies optimal transport; the Kullback–Leibler divergence is *not* symmetric and *not* a metric, even though it is often informally called a "distance".